Centre for Infectious Disease Genomics and One Health
Blusson Hall, Simon Fraser University, Burnaby
www.cidgoh.ca www.imicroseq.ca

iMicroSeq Contextual Data Curation SOP

Version 0.2 Updated March 2026

Disclaimer: As of March 2026 this SOP pertains to contextual data for the initial release of the iMicroSeq Portal only. This is expected to change after the iterative testing phase.

I. Purpose

The purpose of this SOP is to guide the structured curation of contextual data associated with microbial waster-based samples for the iMicroSeq project. This includes both sequencing-based genomics (e.g. whole genome sequencing, metagenomics, amplicon sequencing) and targeted genetic marker approaches (e.g. qPCR assays, 16S/AMR gene detection). Standardising contextual information ensures data is interoperable, shareable, and reusable while also protecting privacy and supporting ethical data use.

This SOP and the supporting materials are available on GitHub at https://github.com/cidgoh/iMicroSeq_Contextual_Data_Specification. These resources have been adapted from the PHA4GE Wastewater Contextual Data Curation SOP and will continue to evolve to encompass additional environmental water sources. This SOP applies to all data contributors submitting contextual data for wastewater-based genomic surveillance activities under the iMicroSeq project.

II. Data

The scope of the standard encompasses contextual data for microbial genomics and genetic marker-based studies derived from water samples, water-associated environmental sites/materials (e.g. biofilm, sediment, aquatic vehicles, wastewater treatment plants, water towers, etc), as well as water-associated hosts (e.g. aquatic wildlife, aquatic plants, etc). Terrestrial samples (environmental sites/materials, hosts) associated with bodies of water can also be included (e.g. wildlife, beach sand, cliff soil, etc). This includes both sequencing-derived data (e.g. genomic characterisation, taxonomic identification, lineage assignment) and targeted assay data (e.g. PCR-based detection, AMR gene identification). Terrestrial samples associated with bodies of water may also be included.

Here data refers to contextual data describing repository accession numbers, sample collection and processing, environmental conditions and measurements, sequencing, bioinformatics and QC metrics, taxonomic identification, AMR detection, lineage/clades, pathogen diagnostic testing, and contributor acknowledgements, as supplied by the data provider. As the standard is intended to create interoperability of data across common domain-specific use cases, not all project-specific data may fit into the data schema of the iMicroSeq Data Portal. Additional data types can be included as broad descriptors

(see “additional data types” field), and/or via sharing URLs linking to protocols, sampling plans, citations, repositories, etc.

Data shared by the iMicroSeq Data Portal can be analyzed in different ways. As such, primary data is considered within scope, however the results of analyses are not included (some exceptions apply e.g. taxonomic mapping for identification of organisms present in a sample is within scope, however quantification/comparative quantification of an organism is excluded).

Note: Different subsets of fields will apply to samples from different contexts (i.e., structured sewage systems vs. contaminated surface waters). Fields not pertinent to the sample type of interest need not be filled. Refer to the Appendix B and the Reference Guide for further instructions regarding different sample types.

III. Key Principles

Use of a standardised template allows for consistent, comparable datasets with easier integration across sites and systems, while also future-proofing for reuse. Collecting contextual data in a standardized way also enables better protection of sensitive data. Consider the following when completing the contextual data:

- **Permission to Share**: Data providers must have the rights and permissions from the data owners to submit and share the contextual data. Ensure alignment with institutional, jurisdictional, and legal data sharing and privacy policies.
- **Privacy and Ethics**: Respect ethical constraints. Use null values when data is not permitted for sharing. Seek guidance from a data steward or privacy officer if needed.
- **Attribution**: Ensure all contributors (sample collection, processing, sequencing, curation) are credited.

IV. Procedure

1. Preparation

- Download the file containing the current iMicroSeq reference guide from the GitHub repository https://github.com/cidgoh/iMicroSeq_Contextual_Data_Specification. We recommend using the DataHarmonizer to curate your data, although you may use any tool suitable if adhering to the iMicroSeq contextual data schema.
- To use the DataHarmonizer as your data collection instrument follow the instructions for use. You can download the latest DataHarmonizer templates at the pathogen genomics package.
- Prior to curating contextual data please:
    - Confirm data sharing permissions with your institution, considering privacy, legal, and ethical constraints.

- Review your dataset accordingly
- Review the fields and terms in the Reference Guide (schema).
- Confirm the data mapping of the contextual data to the iMicroSeq specification with a data steward or supervisor.
- *Note*: *Confirm the level of granularity of information that can be shared publicly and/or privately, with the data steward and/or your privacy officer. The most detailed information allowable should be included here. Different versions (detailed information vs general information) can be stored.*

2. Completing the Template

- Populate the collection template with information from your dataset. You can use the DataHarmonizer for a submission ready template to the iMicroSeq Portal, or alternatively you can use the reference guide to create your own template.
  - Fill in the **required fields** (highlighted yellow in the reference guide, alternatively see the table below).
  - Complete **recommended fields** (highlighted purple in the reference guide, alternatively see the table below) where possible.
  - Complete optional fields (white) if data is available and approved for sharing.
  - Use controlled vocabularies and picklists.
  - Use null values appropriately to indicate reasons for missing data (e.g., not collected, not applicable, restricted access).

Table: Summary of the required (bold) and recommended fields of the iMicroSeq contextual data specification.

Subsection	Required Fields
Database Identifiers	**specimen collector sample ID** sampling site ID sampling event ID INSDC sample accession
Sample Collection & Processing *Note: Evaluate with your supervisor or data steward whether the specimen collector sample ID is considered identifiable by your institutional policies. If not, copy the sample ID into the sample ID field in the validator spreadsheet. If yes, provide the alternative sample ID*	**data steward contact name** **data steward contact email** **organism** **purpose of sampling** **sample collected by organisation name** **sample collection start date** sample collection end date sample collection time duration value sample collection time duration unit **geo_loc_name (country)** **geo_loc_name (state/province/territory)** environmental site

as specified by the lab. Be sure to keep a copy of the key.	environmental material environmental material properties collection device collection method anatomical material anatomical part body product specimen processing endogenous control details
Environmental Conditions & Measurements *Note: Populate only the fields that pertain to your sample. Provide the most granular information allowable according to your organization's data sharing policies. Select the appropriate value from the available pick list (consult the reference guide and curation SOP for further support).*	water catchment area human population measurement value presampling weather conditions sampling weather conditions precipitation measurement value precipitation measurement unit turbidity measurement value turbidity measurement unit fecal contamination indicator fecal contamination value fecal contamination unit
Host information	host (common name) host (scientific name) host age host age unit host age bin
Strain & Isolate Information (only for isolate samples)	**isolate ID** microbiological method
Sequence Information	**purpose of sequencing** assay type **sequenced by organisation name** **sequenced by contact name** **sequenced by contact email** **sequencing instrument** sequencing date sequence submitted by sequencing protocol genomic target enrichment method amplicon pcr primer scheme name
Bioinformatics & QC metrics	raw sequence data processing method dehosting method
Taxonomic Identification Information	read mapping software name read mapping software version taxonomic reference database name taxonomic reference database version

Lineage/clade information	viral type viral subtype

3. Data Handling and Storage

- If IDs or identifiers are anonymised, securely retain the mapping key.

V. Ethical and Practical Considerations

Effective and equitable pathogen surveillance and monitoring via water sampling requires rapid and sustained international collaboration and data sharing. Many of the contextual data elements described in the iMicroSeq contextual data specification are critical for effective public health and environmental decision making and response. However, many of these same elements have ethical, practical, and privacy issues which must be considered before data can be shared. Data governance policies may vary between data types and jurisdictions, thus users of the specification should consult data stewards and privacy officers regarding organization-specific and jurisdiction-specific policies. At the highest level this includes:

- **Granularity**: Share the most granular information allowable, balancing utility with privacy.
- **Geolocation and Temporal Data:** Use accurate and approved spatial/temporal data. Avoid using proxy locations.
- **Methodological Transparency**: Record relevant sampling, processing, and analysis methods. These support data reuse and reproducibility.
- **Licensing and Use**: If applicable, include data use license information.

Below, we highlight a series of common issues and provide suggestions for ways forward. The **Reference Guide** should be consulted for **field-level guidance.**

1. Data Stewardship and Attribution

Data is often generated through the work and contributions of different individuals and organisations (e.g. experimental design, sample collection, microbial isolation, sequencing, laboratory testing/characterization, etc). Attributing the contributions from different partners can be critical for ensuring trust within collaborations, sharing equitable distribution of the benefits (e.g. establishing track records, applying for grants), and for providing accurate contact details for follow-up. As such, data provenance fields such as *sample collected by organisation name*, *isolated by organisation name*, and *sequenced by organisation name* should be filled when possible. Formal organization names (what might appear on official letterhead) should be provided instead of short forms e.g. “University of British Columbia” rather than “UBC”. The *authors* field enables general attribution of different partners involved in data generation, and has no limit.

Data stewardship is the management, protection, and oversight of data assets to ensure quality, security, and accessibility throughout their lifecycle. Individuals can be nominated as the data steward of datasets submitted to the Portal. Data stewards will be responsible for long-term oversight of data, and should have contact information available for follow-up (e.g. *data steward contact name* and *data steward contact email* are required fields).

2. Identifiers and repository accession numbers

When you share information with a public database, you will receive an accession number (a unique identifier in a database enabling the tracking of multiple versions of the data). If you have shared data with a public database, make sure to capture the accession numbers. INSDC members (NCBI, ENA, DDBJ) may provide you with different accession numbers depending on what you share, and how. You can share assemblies and consensus sequences with GenBank (and its equivalents), raw data with Sequence Read Archive (SRA), and contextual data as a BioSample (see reference guide for further information). Information may be organized in BioProjects, and at a higher organizational level, Umbrella BioProjects. Make sure to record all of the applicable accession numbers. Samples, libraries, sites, sequences (raw, processed, consensus etc) and so on can have many identifiers, especially if there is a division of labor or sharing of information across agencies and organizations. The specification has provided fields to capture many of those that are common, but may not capture all of the IDs you require. **It is essential to track IDs of original materials and information** to establish chain-of-custody and for follow-up, if necessary. It is better to track too many IDs than too few. If you require more fields to capture the IDs you need, add them. Some IDs are considered public health identifiable information (PHII). Make sure to check with the appropriate authorities whether the IDs you plan to share are considered identifiable information. If considered identifiable, you may need to create an alternative set of IDs. If you do, make sure to store the key in a safe and secure place.

3. Geographical information

Geographical information (country, province/state/region, city, postal code, latitude/longitude etc) is an essential descriptor of a sampling site. However, when coupled with other contextual data, this information may incidentally reveal the identities of individuals or a community contributing effluent to a given site. Especially for sites with smaller catchment areas, it may be necessary to abstract geographic information before it can be shared. Before sharing data, especially with public repositories, it is important to ensure the data being submitted complies with the permitted level of granularity. Discuss this with the data steward. If sharing latitude and longitude coordinates, do not use the centre of the region or the location of your agency as a proxy, as this implicates

a real location and is misleading. Latitude and longitude should either be shared accurately or kept private.

Sewershed/watershed information can also potentially be used to identify a specific community contributing effluent to a site. For example, if a site is described as a "correctional facility," this could be combined with geographic information to positively identify a specific prison. Proximal sites can also make data identifiable. Especially when a community is smaller or disadvantaged, the potential to identify a specific community should be taken into account when deciding which contextual data to share with collaborators or publicly.

4. Date information

Geographical and temporal information are key elements of infectious disease surveillance programs. Temporal information consists of dates e.g. sample collection date, sample received date, sample sequenced date, etc. This information should be captured using clearly defined event-specific fields (e.g. sample collection date, sample received date, sequencing date), with all dates recorded in ISO 8601 format (YYYY-MM-DD) to ensure consistency and interoperability.

The "sample collection date" represents the time at which the specimen was obtained from its source and should not be inferred or substituted. The "received date" must only be used to record when the sample arrived at the processing laboratory and must not be used as a proxy for collection date, as this introduces systematic bias in temporal analyses. Elements such as "sample collection date" are usually held by the institution that collected the original specimen. As such, you may require permission to acquire this information, or it may be difficult to attain due to other burdens on the data provider (workload, system access, manual curation requirements). If the sample collection date is unavailable, the appropriate null value should be used rather than substituting another date field. This preserves semantic clarity and prevents misinterpretation in downstream analyses.

5. Purpose of sampling/sequencing

Sampling strategy can create biases in the data. A sample may be collected for one purpose, but sequenced for another (e.g. collected for water quality testing, but sequenced for surveillance of circulating lineages and variants). Information about why samples were collected and why they were selected for sequencing (i.e random vs targeted sampling) can help inform hypothesis formation and support decision making, e.g. epidemiological modeling and analyses. Standardized tags are available in the "purpose of sampling" field (e.g. protocol testing, wastewater drug surveillance) and in the "purpose of sequencing" field (e.g. baseline surveillance (random sampling), longitudinal surveillance (repeat sampling)). Free text fields are also available for providing extra information about sampling and the selection of samples for sequencing

called “purpose of sampling details” and “purpose of sequencing details”. A number of standardized phrases are also suggested in the Reference Guide for describing different common surveillance priorities.

6. Methods information

Documenting procedures, bioinformatic processing steps, and quality control (QC) metrics is essential to understanding the context, strengths, and limitations of any analysis. Capturing well-structured methodological information helps to preserve institutional knowledge, supports reproducibility, and ensures that the effort invested in data generation is not lost over time. The iMicroSeq specification includes dedicated fields for experimental design, laboratory protocols, and prioritized metrics. It is strongly recommended that as many of these fields as possible be completed. While the template captures critical water surveillance/monitoring information, it is not intended to function as a Laboratory Information Management System (LIMS). More granular data (e.g., raw concentrations, preparation measurements, detailed run parameters) should be recorded in a LIMS where available. In the absence of a LIMS, the template can serve as the primary record for tracking methodological details.

When completing **bioinformatics fields**, only those relevant to your workflow should be populated. Some fields may appear similar—such as *consensus sequence software name* and *sequence assembly software name*—but they represent distinct methods of analysis. The sequence assembly software field should be completed if your workflow includes an assembly step (for example, de novo assembly), while the consensus sequence software field should be completed if your workflow generates consensus sequences from mapped reads. If both steps occur, both fields should be populated. In cases of uncertainty regarding which fields to complete, the data steward should be consulted prior to submission.

Similarly, when completing the taxonomic information section the relevant fields should be completed if a taxonomic analysis has been conducted. This includes recording the *read mapping software name* and *version*, the *taxonomic reference database name* and *version*, and the *read mapping criteria* applied. Any generated outputs, such as a taxonomic analysis report, should be referenced by filename and associated with the date the analysis was performed.

7. Recording measurements

Where possible, measurements should be recorded as precise numerical values to enable more accurate comparisons across studies and over time. However, it is recognised that in some cases only value ranges (bins) may be available due to data collection methods, reporting requirements, or privacy considerations.

The iMicroSeq template is designed to accommodate both exact and binned measurements, allowing for flexible data capture across laboratories with varying public health priorities, capacities, and reporting needs. Exact values and binned ranges should be recorded in their respective fields (e.g., *water catchment area human population measurement value* vs *water catchment area human population range*).

With the exception of individual count measurements, all measurements should follow a structured value, unit, and method pattern of capture where:

- **Value** represents the quantitative observation (either a precise number or a defined bin), forming the core data used in analysis.
- **Unit** specifies the scale and standard of measurement (e.g. CFU/mL, copies/L), enabling harmonisation, comparability, and correct interpretation across datasets.
- **Method** describes how the measurement was generated (e.g. assay type, instrument, protocol), providing essential context for assessing data quality, comparability, and potential bias.

Capturing these components in separate but linked fields preserves semantic clarity, supports ontology alignment, and enables computational reuse.

8. Genetic marker information

Genetic marker information should be captured using the diagnostic and assay specific fields when a study targets the presence, absence or quantity of specific genes, genomic regions or variants (e.g. PCR assays, AMR gene detection). These approaches differ from genomics in that they focus on predefined targets rather than generating whole genome or metagenomic sequence data.

For each marker, the *target taxonomic name* should describe the taxon of interest (if applicable) using the NCBITaxon ontology to capture the scientific name, while *assay target name* and *gene symbol* capture specific region or gene being interrogated (e.g. AMR genes such as gyrA, or PCR targets). *Assay target name* should be used when a region of a gene is being specified, or if the target of amplification is not a gene (e.g. untranslated region (UTR)). Results should follow the structured value–unit–method (see above) pattern using *diagnostic target presence*, *diagnostic measurement value* (e.g. Ct), *diagnostic measurement unit*, and *diagnostic measurement method* (e.g. qPCR). If a study is based solely on genetic markers, genomics fields (i.e. sequencing information) do not need to be completed. However if both sequencing-based genomics and marker-based assays are performed, both sets of fields should be populated to support integrated interpretation.

9. Normalization

Primary data (e.g. experimental measurements) can be normalized in many different ways. It is best to provide raw data and include indicators, endogenous control information, and normalization metrics/parameters separately - as much as possible. This separation enables data users to refactor variables more easily for comparison with different studies and methods. Fields such as *experimental control details* and *endogenous control details* can be used to describe normalization methods.

10. Organism, host and target taxonomic names

The *organism*, *host name (scientific name)*, and *target taxonomic name* fields capture related but distinct aspects of biological context and should be completed consistently for semantic clarity and correct interpretation of the data. The *organism* field describes the microorganism(s) identified in the sample (i.e. the subject of the analysis), recorded at the **lowest appropriate taxonomic level** where possible. The *host name (scientific name)*, describes the organism from which the microbial sample was collected (e.g. *Homo sapiens*) and is populated only for host-associated samples. The *target taxonomic name* field is linked to diagnostic or assay based data (e.g. PCR or AMR), and defines the organism the assay is designed to detect (this should be an NCBITaxon compliant name where possible). In practice, *organism* reflects what is observed in the sample, *host name* reflects the sampling context (if applicable), and *target taxonomic name* reflects what was tested for.

11. Environmental sites vs materials

In this specification, an *environmental material* is considered to be a substance or movable object, whereas an *environmental site* is considered to be a location or a fixture of a built/natural environment. The boundary between these elements can be fuzzy in certain situations. Data modelling exercises by standards developers help place new terms in the most appropriate category, and picklists are provided to streamline data classifications for data providers/users. If the desired term is not found in one picklist menu, the other picklist menu should be searched.

12. Anatomical parts, materials, and body products

Different samples may be taken from a host, such as tissue (or other anatomical materials such as blood) from a particular anatomical part. In this specification, an *anatomical part* is an anatomical location/structure (e.g. lung, knee, cloaca), whereas an *anatomical material* (e.g. blood, tissue, cecal contents) is material that can be removed from an *anatomical part*. A *body product* is a material meant to be excreted/secreted from a host, and is usually a waste product (e.g. feces, urine, vomit).

13. Multi-tagging

"Multi-tagging" generally refers to the use of multiple labels to a contextual data record. Many of the fields enable multiple values to be added from a picklist separated by a semi-colon. These formats are compatible with public repository requirements, such as the National Center for Biotechnology Information (NCBI). Multiple keywords (e.g. software tools) can also be included in free text fields separated by a semi-colon.

14. Available data types

The iMicroSeq Data Portal is intended to support the storage and sharing of datasets generated to answer a wide variety of questions, across different sectors and domains of knowledge, using a variety of methods. Often, experiments include the production of data that characterizes organisms and samples beyond genomic and genetic attributes. Due to the generalist nature of the Data Portal, the iMicroSeq data specification prioritizes the most common data types and data elements across a range of water-based use cases. This prioritization increases the types of genomic and genetic datasets that can be included in the database, however, it may limit the range of additional data types that can be included (e.g. chemical profiles that differ according to agency/standard/water type, mollusc shell lengths, land use information, antibiotic administration, conductivity measurements, watershed shapefile details, etc). To enable data providers to flag the availability of additional data types and to encourage collaborations, the *additional data types* field provides a picklist of data tags that can be used to raise awareness of additional project-specific data. Data users can follow-up with the data steward for more information.

VI. SOP Provenance and Future Expansion

This SOP is based on the PHA4GE Wastewater Specification, developed collaboratively across international partners. iMicroSeq will continue to align with global best practices in public health and One Health data standards. Contributors are encouraged to consult the PHA4GE GitHub repository and preprint for detailed guidance and updates:

- **GitHub**: https://github.com/pha4ge/Wastewater_Contextual_Data_Specification
- **Preprint**: https://osf.io/preprints/osf/z79vk_v1

This SOP will be updated as the project expands to include additional water types.

Contact

For template access, guidance, or questions, please contact Emma Griffiths at emma_griffiths@sfu.ca or add an issue on the iMicroSeq GitHub repository.

Version	Date	Contributor	Log
0.1	Aug 2025	Charlie Barclay	Initial draft
0.2	Mar 2026	Emma Griffiths, Charlie Barclay	Updated required and recommended, and relevant links